EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Runs always equal number of unique values.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

# Unfamiliar programming challenges

Name: _____ Class: _____ Date: _____

**1. What is a loop invariant?**

**2. What distinguishes a run from a distinct value?**

**3. Why handle empty arrays explicitly?**

**4. For a=[1,1,2,1], give count of distinct values and runs.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

A property true before and after each completed iteration.

### 2. Model answer

Runs count contiguous groups; distinct counts globally unique values.

### 3. Model answer

Otherwise a[0] could be invalid and initial count wrong.

### 4. Model answer

Distinct values 2; runs 3 ([1,1],[2],[1]).

### By the end, you can…

- A run is a maximal group of equal consecutive values, not a unique value across the whole array.
- Maintain invariant: after index i, the count equals the number of runs in the processed prefix.
- Handle empty input and test all-equal, alternating and separated-repeat arrays.

### Vocabulary — know these terms

run — Consecutive equal-value group.

invariant — Property maintained across iterations.

edge case — Boundary or unusual input.

distinct — Different values irrespective of position.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

For [1,2,1], compare runs and distinct counts, and assess time/extra-space complexity of CountRuns. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Trace CountRuns for [3,3,4,4,3,5,5]. State where runs increment. [6]

### Question 2

Write a valid C# CountRuns(int[] a) returning runs and handling empty input. [8]





NEW LEARNING

# Read, annotate and explain

### Understand

A run is a maximal group of equal consecutive values, not a unique value across the whole array.

### Apply

Maintain invariant: after index i, the count equals the number of runs in the processed prefix.

### Evaluate

Handle empty input and test all-equal, alternating and separated-repeat arrays.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Trace CountRuns on [4,4,5,5,5].

### Predict first

Consecutive repeated fives belong to the same second run.

### Explain and check

Explain the reason for the result.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve